Byounghwa Lee

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ORCID Google Scholar LinkedIn

Professional Experience

Electronics and Telecommunications Research Institute (ETRI), Senior Researcher — Daejeon, KR
Integrated Intelligence Research Section (Jan 2024–Present) — Aug 2021 – Present
Cyberbrain Research Section (Aug 2021–Dec 2023)

- Scientific representation learning: science-specific sequence encoders and tokenizers; protein–ligand representations (2026–Present).
- Speech-based cognitive assessment: multimodal and graph learning, MCI classification models, and field deployment (2024–2026).
- Continual and self-adaptive AI: temporal point processes, burst dynamics, graph methods, and explainable models (2021–2023).

Samsung Research (Samsung Electronics), Staff Engineer — Seoul, KR
Global AI Center, Big Data Team, Data Analytics Lab — Mar 2019 – Jul 2021

- Cross-Domain Recommendation Model Using Self-Supervised Learning (2021)
  Keywords: Recommender Systems, Cross-Domain Recommendation, User Behavior Modeling, Self-Supervised Learning
- Multi-Touch Attribution Model Using AI (2020)
  Keywords: Online Advertising, Customer Journey Model, Multi-Touch Attribution, Self-Attention
- Inventory Optimization Using AI (2019)
  Keywords: Demand Forecasting, Inventory Optimization, Time Series Forecasting

Education

Ph.D. **Pohang University of Science and Technology (POSTECH)**, Physics *Combined M.S./Ph.D.* — Pohang, KR, Mar 2012 – Feb 2019

- Advisor: Woo-Sung Jung and Hang-Hyun Jo
- Research Topics: Statistical Physics, Complex Networks, Bursty Dynamics
- Dissertation: Generative Model and Method for Correlated Bursty Dynamics

B.S. **Pohang University of Science and Technology (POSTECH)**, Physics — Pohang, KR, Mar 2007 – Feb 2012

- Undergraduate Research Program: Single Molecule Fluorescence Study on Highly Bent dsDNA (2011)

Research Interests

AI for Science and Complex Systems: statistical physics; scientific representation learning; structure–dynamics relations; model validation.

Graph Learning and Constrained Generation: complex networks; graph representations; constraint-aware generation; learning and search.

Event-Sequence Learning: temporal point processes; bursty dynamics; memory and temporal dependence; probabilistic prediction.

Data-Efficient Learning and Scientific Discovery: self-supervised and transfer learning; generalization; uncertainty-aware sequential experimental design.

Publications prior to 2020 were published under the name Byoung-Hwa Lee.

Publications

Simultaneous Multi-Task Assessment of Five Neurological Language Disorders via Cross-Corpus Multimodal Learning — Oct 2026

Byounghwa Lee, Hwa Jeon Song, Byung Ok Kang

Scientific Reports 16 (2026), published online

10.1038/s41598-026-76495-9

Multimodal Alzheimer's Disease Recognition from Image, Text and Audio — Aug 2025

Byounghwa Lee, Hwa Jeon Song, Young-Jin Park, Byung Ok Kang

Scientific Reports 15 (1), 29038

10.1038/s41598-025-14998-7

Alzheimer's Disease Recognition Using Graph Neural Network by Leveraging Image-Text Similarity from Vision Language Model — Jan 2025

Byounghwa Lee, Jeong-Uk Bang, Hwa Jeon Song, Byung Ok Kang

Scientific Reports 15 (1), 997

10.1038/s41598-024-82597-z

Burst and Memory-aware Transformer: Capturing Temporal Heterogeneity — Dec 2023

Byounghwa Lee, Jung-Hoon Lee, Sungyup Lee, Cheol Ho Kim

Frontiers in Computational Neuroscience 17, 1292842

10.3389/fncom.2023.1292842

Latent-Variable Classifiers Based on Total Correlation for Dynamic Environments — Oct 2023

Cheol Ho Kim, **Byounghwa Lee**, Jung-Hoon Lee, Ock Kee Baek

Proceedings of the 14th International Conference on Information and Communication Technology Convergence (ICTC)

10.1109/ICTC58733.2023.10393699

Explainable ML Models for a Deeper Insight on Treatment Decision for Localized Prostate Cancer — Jul 2023

Jang Hee Han, Sungyup Lee, **Byounghwa Lee**, Ock-kee Baek, Samuel L. Washington III, Annika Herlemann, Peter E. Lonergan, Peter R. Carroll, Chang Wook Jeong, Matthew R. Cooperberg

Scientific Reports 13 (1), 11532

10.1038/s41598-023-38162-1

Brain-Inspired Predictive Coding Improves the Performance of Machine Challenging Tasks — Nov 2022

Jangho Lee, Jeonghee Jo, **Byounghwa Lee**, Jung-Hoon Lee, Sungroh Yoon

Frontiers in Computational Neuroscience 16, 1062678

10.3389/fncom.2022.1062678

Copula-Based Algorithm for Generating Bursty Time Series — Aug 2019

Hang-Hyun Jo, **Byoung-Hwa Lee**, Takayuki Hiraoka, Woo-Sung Jung

Physical Review E 100 (2), 022307

10.1103/PhysRevE.100.022307

Hierarchical Burst Model for Complex Bursty Dynamics — Aug 2018

Byoung-Hwa Lee, Woo-Sung Jung, Hang-Hyun Jo

Physical Review E 98 (2), 022316

10.1103/PhysRevE.98.022316

Analysis on the Urban Street Network of Korea: Connections between Topology and Meta-Information — May 2018

Byoung-Hwa Lee, Woo-Sung Jung

Physica A: Statistical Mechanics and its Applications 497, 15–25

10.1016/j.physa.2017.12.131

Presentations

International Conferences

Byung Ok Kang, **Byounghwa Lee**, Jeong-Uk Bang, Hwa Jeon Song, Young Jin Park, "Intelligent System for Early Detection of Mild Cognitive Impairment and Dementia Risk Through Spoken Language Analysis" *2025 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), Show & Tell* — Apr 11, 2025

Byoung-Hwa Lee, Woo-Sung Jung, "Analyzing the Urban Street Networks of Korea" *NetSci 2016* — Jun 1-3, 2016

Byoung-Hwa Lee, Woo-Sung Jung, "Centrality in Urban Street Networks of Korea: From Classification to Model" *International Conference on Social Modeling and Simulations & Econophysics Colloquium* — Nov 4, 2014

Byoung-Hwa Lee, Inho Hong, Woo-Sung Jung, "Complex Network Analysis of the Korean Transportation Network" *The 15th Asia Pacific Industrial Engineering and Management Systems Conference (APIEMS)* — Oct 12-15, 2014

Byoung-Hwa Lee, Inho Hong, Woo-Sung Jung, Okyu Kwon, "Statistical Properties of the Korean Transportation Network as a Complex Network" *European Conference on Complex Systems (ECCS) '14* — Sep 23, 2014

Byoung-Hwa Lee, Woo-Sung Jung, "Prediction of Congestion Sites in Pohang" *Econophysics Colloquium 2013 & Asia Pacific Econophysics Conference 2013* — Jul 30, 2013

International Workshops

Byoung-Hwa Lee, Woo-Sung Jung, "Complex Network Analysis of the Urban Street Structure of Korean Cities" *APCTP 2016 Workshop on Frontiers of Physics: Push the Envelope of Statistical Physics: Econo, Social, Bio and Beyond* — Dec 14, 2016

Inho Hong, **Byoung-Hwa Lee**, Hang-Hyun Jo, Woo-Sung Jung, Hyejin Youn, "Industrial Dynamics in Urban Areas" *Application of Econophysics and Social Physics: Winter Workshop* — Feb 22-23, 2016

Byoung-Hwa Lee, Woo-Sung Jung, "Classification of the Street Patterns by Using Centrality Analysis" *Physics of Social Complexity* — Nov 2-4, 2015

Byoung-Hwa Lee, "Roads Network in Pohang" *The 5th International Symposium on IT Convergence Engineering* — Jul 11-12, 2013

Domestic Conferences

Byoung-Hwa Lee, Hang-Hyun Jo, "Hierarchical Modeling for Correlated Bursty Dynamics" *2018 Korean Physical Society Spring Meeting* — Apr 25, 2018

Byoung-Hwa Lee, Hang-Hyun Jo, "Hierarchical Model for Correlated Bursty Dynamics" *The 19th Workshop for Statistical Physics* — Aug 28-30, 2017

Byoung-Hwa Lee, Woo-Sung Jung, "Complex Network Analysis of the Korean Urban Street Structure" *2015 Korean Physical Society Spring Meeting* — Apr 22, 2015

Byoung-Hwa Lee, Woo-Sung Jung, "Centrality Analysis in Street Networks of Korea: From Classification to Model" *2014 Korean Physical Society Fall Meeting* — Oct 23, 2014

Byoung-Hwa Lee, Woo-Sung Jung, "Centrality Analysis in the Korean Street Networks: From Classification to Model" *Application of Econophysics and Social Physics: Summer Workshop* — Aug 22, 2014

Byoung-Hwa Lee, Woo-Sung Jung, "Analysis of Road Networks Using Complex Network: Predicting Traffic Congestion Areas in Pohang and Changwon via Betweenness Centrality" *The 8th Conference on Complexity* (in Korean) — Nov 16, 2013

Byoung-Hwa Lee, Woo-Sung Jung, "Investigation of Road Networks in Two Korean Cities: Pohang and Changwon" *The 17th Workshop for Statistical Physics* — Nov 2, 2013

Byoung-Hwa Lee, Woo-Sung Jung, "Investigation of Road Networks in Two Korean Cities: Pohang and Changwon" *2013 Korean Physical Society Fall Meeting* — Oct 31, 2013

Academic Service and Recognition

Journal Reviewer

Alzheimer's & Dementia: Diagnosis, Assessment & Disease Monitoring	2025	International Journal of Machine Learning and Cybernetics	2026
Data & Policy	2026	Journal of King Saud University – Computer and Information Sciences	2026
Discover Artificial Intelligence	2026	Multimedia Systems	2025
Discover Internet of Things	2026	Nonlinear Dynamics	2025
IEEE Access	2025–2026	Scientific Reports	2025
IET Biometrics	2026	The Journal of Supercomputing	2025–2026
Journal of Big Data	2025	Wireless Networks	2025–2026

Recognition: IEEE Access Exceptional Reviewer 2025

Visiting Experience

Aalto University, Department of Biomedical Engineering and Computational Science — Helsinki, Finland, Feb 2015

Teaching Experience

Department of Physics, POSTECH, Teaching Assistant — Spring 2013
Statistical Mechanics

Department of Physics, POSTECH, Teaching Assistant — Fall 2012
General Physics II

Department of Physics, POSTECH, Teaching Assistant — Spring 2012
General Physics I

Research and Technology Contributions

Speech-Based MCI Screening — 2024–2026
Primary developer of the classification model and lead contributor to field deployment; contributed to technology transfer.

Digital-Human System Integration — 2026
Developed the user interface, real-time voice and session handling, and server integration; managed system operation.

Research Resources

Korean Kang Corpus, DementiaBank/TalkBank
Contributor to a Korean speech corpus for cognitive assessment.
Corpus page 10.21415/PJVS-9X49

Patents – Republic of Korea

Granted Patents
Issued after examination. Role follows the inventor order in the patent record.

[KR-G1] Device and Method for Generating Networks Using Complex Network Properties
First inventor; **KR 10-3005917**; granted Aug 12, 2026.
Application 10-2022-0001721; filed Jan 5, 2022. Assignee: ETRI.

[KR-G2] Method and Apparatus for Predicting Degenerative Brain Function Decline and Cognitive Impairment Using a Vision-Language Model and Graph Neural Network
First inventor; **KR 10-2968743**; granted May 20, 2026.
Application 10-2025-0078633; filed Jun 16, 2025. Assignee: ETRI.
Earlier priority application: 10-2024-0079454 (Jun 19, 2024).

[KR-G3] Electronic Apparatus and Control Method Thereof
First inventor; **KR 10-2900832**; granted Dec 16, 2025.
Application 10-2021-0004603; filed Jan 13, 2021. Assignee: Samsung Electronics.

Patent Applications
Filed and not listed here as granted. Assignee: ETRI.

[KR-A1] Apparatus and Method for Predicting Emotions and Providing Real-Time and Post-Response Support Using an End-to-End Model Integrated with a Foundation Model
Co-inventor; application **KR 10-2026-0030997**; filed Feb 19, 2026.

[KR-A2] Method and Apparatus for Building a Multimodal Foundation Model for Predicting Various Neurological Disorders
First inventor; application **KR 10-2026-0021243**; filed Feb 3, 2026.

[KR-A3] Apparatus and Method for Reproducing Tabular Data
Co-inventor; application **KR 10-2024-0091725**; filed Jul 11, 2024.
Publication: KR20260009569A.

[KR-A4] Apparatus and Method for Building an Integrated Medical-Data Model Using Incremental Learning
Co-inventor; application **KR 10-2024-0058637**; filed May 2, 2024.
Publication: KR20250159481A.

[KR-A5] Method of Forecasting Time Series Data
First inventor; application **KR 10-2024-0055564**; filed Apr 25, 2024.
Publication: KR20250156492A.

[KR-A6] Machine Learning Device Using Adaptive Feature Selection and Method of Operating the Same
Co-inventor; application **KR 10-2023-0116366**; filed Sep 1, 2023.
Publication: KR20250033794A.

[KR-A7] Time Series Pattern Prediction Device and Method of Operating the Same
First inventor; application **KR 10-2023-0101823**; filed Aug 3, 2023.
Publication: KR20250020221A.

Patent Applications – United States

Filed applications; no U.S. grant is claimed. Assignee: ETRI unless otherwise indicated.

[US-A1] Method and Apparatus for Predicting Degenerative Brain Function Decline and Cognitive Impairment Using a Vision Language Model and Graph Neural Network
First inventor; application **US 19/242,069**; filed Jun 18, 2025.
Publication: US20250316388A1.

[US-A2] Apparatus and Method for Reproducing Tabular Data
Co-inventor; application **US 18/918,943**; filed Oct 17, 2024.
Publication: US20260017529A1.

[US-A3] Machine Learning Device Using Adaptive Feature Selection and Method of Operating the Same
Co-inventor; application **US 18/667,206**; filed May 17, 2024.
Publication: US20250077966A1.

[US-A4] Time Series Pattern Prediction Device and Method of Operating the Same
First inventor; application **US 18/602,429**; filed Mar 12, 2024.
Publication: US20250045571A1.

Earlier Applications – Discontinued or Abandoned

[US-H1] Device and Method for Generating a Network Using Complex Network Properties
First inventor; application **US 17/936,254**; filed Sep 28, 2022.
Publication: US20230214672A1.
Prosecution discontinued by assignee (2026); no grant claimed.

[US-H2] Electronic Apparatus and Control Method Thereof
First inventor; application **US 17/668,993**; filed Feb 10, 2022.
Publication: US20220222716A1. Assignee: Samsung Electronics.
Abandoned; no grant.
Related PCT application: PCT/KR2021/008634 (WO2022154199A1).